SC306 Homework 4

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Question 1

Creation of birth weight table

```
bw = matrix(c(110, 1073, 289, 15896), nrow=2, byrow = TRUE)
colnames(bw) = c('Low', 'Normal')
rownames(bw) = c('Cocaine', 'No Cocaine')
knitr::kable(bw)
```

	Low	Normal
Cocaine	110	1073
No Cocaine	289	15896

A) The incidence measure we can calculate given the data is incidence proportion. $IP = \frac{\text{Total number of new cases during interval}}{\text{Total population at risk}}$.

B) $IP_{Cocaine} = \frac{110}{110+289} = 0.276$

C) $IP_{Non-Cocaine} = \frac{1073}{1073+15896} = 0.0063$

D) The relative risk of low birth weight is $\frac{0.276}{0.063} = 4.38$.

E) A relative risk of 4.38 means mothers who used Cocaine during pregnancy had 4.38 times the risk of having a baby with low birth weight compared to mothers who did not use Cocaine during pregnancy.

F) Birth weight table

```
(riskratio(bw, rev = "b"))
```

```
## $data
##            Normal Low Total
## No Cocaine  15896 289 16185
## Cocaine      1073 110  1183
## Total       16969 399 17368
##
## $measure
```

```
##                          NA
## risk ratio with 95% C.I. estimate   lower    upper
##               No Cocaine 1.000000      NA       NA
##               Cocaine    5.207422 4.21475 6.433893
##
## $p.value
##             NA
## two-sided    midp.exact fisher.exact   chi.square
##   No Cocaine         NA           NA           NA
##   Cocaine             0 1.500218e-38 3.033658e-62
##
## $correction
## [1] FALSE
##
## attr(,"method")
## [1] "Unconditional MLE & normal approximation (Wald) CI"
```

The risk ratio is 5.21 with a 95% CI of [4.21 : 6.43]. The interpretation for the 95% confidence interval is that given our analysis of the data, we are 95% confident that the true risk ratio is between 4.21 and 6.43. Cocaine use is significantly associated with low birth weight, as shown by the value lower bound of our confidence interval considerably exceeding 1.

Question 2

```r
IPR.conf = function(a, b, c, d, conf) {
  ipr.hat = (a/(a+b)) / (c/(c+d))                  # Predicted IPR
  alpha = 1 - conf
  z = qnorm(1-alpha/2)

  var.ln.ipr = (b/(a*(a+b))) + (d/(c*(c+d)))       # Natural log of IPR variance
  se = sqrt(var.ln.ipr)                            # Standard error

  ln.lower = log(ipr.hat) - z * sqrt(var.ln.ipr)  # Upper/Lower Log Bounds
  ln.upper = log(ipr.hat) + z * sqrt(var.ln.ipr)

  lower = exp(ln.lower)                            # Upper/Lower Bounds
  upper = exp(ln.upper)

  data.frame(
    IPR = ipr.hat,
    "Lower Bound" = lower,
    "Upper Bound" = upper,
    "Confidence Interval" = conf,
    check.names = F
  )}
kable(IPR.conf(110, 1073, 289, 15896, .95), digits = 3)
```

IPR	Lower Bound	Upper Bound	Confidence Interval
5.207	4.215	6.434	0.95

The new interval is effectively identical to the interval from Problem 1.

Question 3: BMI and Mortality

A) The exposure factor is BMI category.

B) The outcome is all-cause mortality.

C) Incidence density can be computed using this data because the follow-up is measured in person-years.

D) Completed table

BMI	Follow up (person-years)	Number of deaths	Incidence of mortality	Relative Risk
Normal	5995	19	0.00317	1.00
Underweight	8479	71	0.00837	2.64
Overweight	1562	7	0.00448	1.41
All	16036	97	0.00605	NA

E) The underweight population are at greater risk of mortality than the overweight population. The underweight population has an all cause mortality rate 2.64 times higher than normal, while the overweight population has an all cause mortality rate 1.41 times higher than normal.

Question 4: Clinical Epidemiology

A) Specificity $= \frac{460}{460+40} = 0.92$

B) Sensitivity $= \frac{840}{840+160} = 0.84$

C) The positive predictive value of this test is: $\frac{840}{840+40} = 0.955$

D) The negative predictive value of this test is $\frac{460}{460+160} = 0.742$

Question 5

A) Because this test is not inflated by a disproportionate number of responses in one category, we use the percent agreement formula: $\frac{A+D}{A+B+C+D}$.

Percent agreement here is $\frac{110+61}{110+30+5+61} = 0.830$

B) $\kappa = \frac{\text{\% agreement - \% agreement due to chance}}{\text{100\% - \% agreement due to chance}}$

Expected in High, High: $\left(\frac{91}{206}\right)\left(\frac{66}{206}\right) \times 206 = 29.2$

Expected in Low, Low: $\left(\frac{115}{206}\right)\left(\frac{140}{206}\right) \times 206 = 78.2$

Expected agreement proportion: $\frac{29.2+78.2}{206} = 0.521$

Observed agreement: 0.830 (from before).

Then, $\kappa = \frac{0.830-0.521}{1-0.521} = 0.645$.

Question 6

Total population: $303 + 162 + 354 + 290 = 1109$

A) The prevalence of BCC is $\frac{303+354}{1109} = 0.592$ or 59.2%

B) No, because this is a case-control study, so a non-representative population exists because people with BCC are intentionally selected for the study.

C) The odds ratio is the odds of exposure among cases divided by the odds of exposure of the controls. $OR = \frac{303/354}{162/290} = 1.53$

D) The odds ratio means the odds of tanning bed use among those with Basal Cell Carcinoma are 1.53 times higher than those without Basal Cell Carcinoma.

E) $OR_{disease} = \frac{303/162}{354/290} = 1.53$ The odds ratio given disease is 1.53.

F) They are the exact same value. The odds ratio is the same if you calculate odds of disease given exposure vs. odds of exposure given disease.

G) No, since this is a case-control study, the number of cases and controls observed was a deliberate choice by those conducting the study, not numbers representative of the general population.